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Diabetes and cancer: therapeutic implications

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Dear Dr. Lipshultz,

Please find attached a revised version of our manuscript entitled "*Diabetes* and Cancer: Therapeutic Implications" which we would like to resubmit for consideration in *Cardio-Oncology*.

Your comments and those of the reviewers were highly insightful and enabled us to greatly improve the quality of our manuscript. In the following pages are our point-by-point responses to each of the comments of the reviewers.

In accordance with journal policy, we provide competing interests for Dr. Javed Butler, a coauthor, who is a consultant to Abbott, Adaptyx, American Regent, Amgen, Applied Therapeutic, AskBio, Astellas, AstraZeneca, Bayer, Boehringer Ingelheim, Boston Scientific, Bristol Myers Squibb, Cardiac Dimension, Cardior, CSL Behring, CVRx, Cytokinetics, Edwards, Element Science, Faraday, Foundry, Innolife, Impulse Dynamics, Imbria, Inventiva, Ionis, Lexicon, Lilly, LivaNova, Janssen, Medtronic, Merck, Occlutech, Owkin, Novartis, Novo Nordisk, Pfizer, Pharmacosmos, Pharmain, Prolaio, Regeneron, Renibus, Roche, Salamandra, Sanofi, SC Pharma, Secretome, Sequana, SQ Innovation, Tenex, Tricog, Ultromics, Vifor, and Zoll. No other authors report competing interests.

All authors have approved the manuscript for submission. The content of the manuscript has not been published or submitted for publication elsewhere.

We hope that the revisions in the manuscript and our accompanying responses will be sufficient to make our manuscript suitable for publication in Cardio-Oncology.

Best regards,

Stavros Fay

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Diabetes and Cancer: Therapeutic Implications

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ABSTRACT

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Background

The growing coexistence of diabetes and cancer contributes substantially to global morbidity and mortality. Beyond the independent burden of each disease, their intersection introduces unique challenges, particularly in cardiovascular risk. Patients with diabetes, especially those with underlying diabetic cardiomyopathy (DbCM), are at increased risk for cancer therapy-related cardiac dysfunction (CTRCD). Yet, the interplay between DbCM and CTRCD remains underexplored despite clear evidence of overlapping mechanisms and heightened vulnerability in this population.

Main Text

DbCM and CTRCD share key pathophysiologic features, including oxidative stress, mitochondrial dysfunction, inflammation, fibrosis, and impaired calcium handling, which collectively reduce myocardial reserve during cancer therapy. Even in patients without overt DbCM, diabetes management complicates cancer care by influencing treatment tolerance and outcomes. Antidiabetic agents exert pleiotropic effects on both cardiac and oncologic pathways: insulin may worsen remodeling and potentially amplify tumorigenic signaling, whereas metformin and sodium-glucose cotransporter 2 inhibitors (SGLT2i) demonstrate promise for cardioprotection and modulation of cancer therapy responses. This review synthesizes the current evidence on shared pathobiology, highlights drug-drug interactions that may exacerbate cardiotoxicity and discusses therapeutic opportunities that could improve patient outcomes.

Conclusions

Understanding the intersection of diabetes and cancer is essential for reducing cardiac complications in this high-risk population. Integrated management strategies should prioritize early risk stratification, interdisciplinary collaboration, and judicious selection of antidiabetic therapies with cardioprotective potential. Research gaps remain in defining optimal monitoring protocols, refining pharmacologic strategies, and developing tailored guidelines. Addressing

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these gaps is critical to advancing cardio-oncology care and improving survivorship for patients living with both diabetes and cancer.

Keywords: Diabetic Cardiomyopathy (DbCM); Cancer Therapy-Related Cardiac Dysfunction (CTRCD); Cardio-oncology; Cardiotoxicity; Diabetes; Cancer; Pharmacologic Interactions; Insulin; Adiposity; Sodium-glucose cotransporter-2 (SGLT2) inhibitor (SGLT2i)

Background

Diabetes and cancer contribute significantly to morbidity, mortality, and healthcare burden globally [1-5], and a growing number of patients are living with both conditions [6]. Patients with diabetes are considered to be particularly at high risk for the development of cancer therapy related to cardiac toxicity with a spectrum of cancer therapeutics, especially those with diabetic cardiomyopathy (DbCM). Diabetes therapies may predispose patients with cancer to increased cardiac complications. Conversely, through complex mechanisms, other diabetes therapies might be clinically beneficial in this patient population [7].

Comorbid diabetes affects the care of patients with cancer, particularly as it relates to cardiovascular complications [8]. Cancer therapy-related cardiac dysfunction (CTRCD) has emerged as a major focus in oncology [9], and many cancer therapies are associated with cardiotoxic effects [10-13]. CTRCD, typically defined as a decline in left ventricular (LV) systolic function, limits the ability to complete potentially curative cancer therapy and contributes substantially to morbidity and mortality during treatment and survivorship [11,14,15]. The presence of traditional cardiac risk factors further amplifies the risk of potential cardiac toxicities [16,17].

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Diabetes is associated with increased cardiovascular risk [18,19] and may exert direct effects on myocardial structure and function, with subsequent development of DbCM [20-23]. The presence of DbCM predisposes patients to myocardial vulnerability when additional stressors such as potentially cardiotoxic cancer therapies are introduced [24, 25]. Both DbCM and CTRCD involve shared pathways [26-34].

Glucose-lowering agents such as metformin, sodium-glucose cotransporter 2 (SGLT2) inhibitors (SGLT2i), and glucagon-like peptide-1 (GLP-1) receptor agonists (GLP-1 RAs) exert pleiotropic effects that influence pathways involved in cell growth, angiogenesis, inflammation, and myocardial remodeling [35, 36] (Table 1). Conversely, insulin and sulfonylureas may promote tumor progression or interfere with metabolic stress responses during therapy [37-40]. This review synthesizes current knowledge regarding the interactions between diabetes and cancer therapies, highlighting areas of synergy and conflict between metabolic and oncologic pharmacotherapy, explores shared pathophysiology between DbCM and CTRCD, and identifies key gaps in knowledge that must be addressed to optimize care for this growing population of high-risk patients (Figure 1).

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- Diabetes and cancer frequently co-occur, amplifying cardiovascular risk and increasing vulnerability to CTRCD, especially in those with DbCM
- Both DbCM and CTRCD share overlapping pathophysiologic mechanisms, and diabetes therapies can either worsen or mitigate cardiac complications in cancer patients.
- This review examines the interplay between diabetes, cancer therapies, and cardiotoxicity, highlighting therapeutic synergies, conflicts, and critical gaps in current management strategies.

PATHOPHYSIOLOGIC OVERLAP BETWEEN DbCM AND CTRCD

Diabetic Cardiomyopathy

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DbCM is defined as myocardial dysfunction in patients with diabetes in the absence of overt coronary disease, valvular disease, or hypertension [41]. Often subclinical in early stages, DbCM leads to diastolic dysfunction, impaired ventricular compliance, and eventually to clinical heart failure (HF) [42, 43]. DbCM is characterized histologically by interstitial and perivascular fibrosis, myocardial lipid accumulation, and microvascular rarefaction [44, 45]. At the molecular level, it is driven by a constellation of processes including hyperglycemia, insulin resistance, mitochondrial dysfunction, increased reactive oxygen species (ROS) generation, and activation of profibrotic signaling cascades such as the transforming growth factor-beta (TGF-β) and nuclear factor-kappa B (NF-κB) pathways [24, 46-51]. Inflammation, oxidative stress, advanced glycation end products (AGEs), and abnormal calcium handling all contribute to myocardial stiffening, altered conduction, and apoptosis [52-55].

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Cancer Therapy-Related Cardiac Dysfunction

CTRCD represents a heterogeneous spectrum of myocardial injury arising from exposure to anticancer agents [9]. Although diverse in presentation, these toxicities share overlapping molecular and pathophysiologic pathways that impair myocardial contractility and reserve. Among the most extensively studied are anthracyclines, which produce cumulative, dose-dependent, and often irreversible cardiomyopathy through oxidative stress, mitochondrial injury, and interference with topoisomerase IIβ [56-59]. Human epidermal growth factor receptor 2 (HER2)-targeted therapies such as trastuzumab typically induce a reversible decline in cardiac function, reflecting transient disruption of HER2-mediated survival signaling within cardiomyocytes [60-62]. Immune checkpoint inhibitors (ICIs), which enhance antitumor immunity by blocking inhibitory pathways such as programmed death-ligand 1 (PD-L1) and cytotoxic T-lymphocyte antigen 4 (CTLA-4), may lead to myocarditis or subclinical inflammatory injury through aberrant activation of cytotoxic T cells [63], whereas tyrosine kinase inhibitors (TKIs) impair mitochondrial energetics and promote apoptotic cell death by disrupting pro-survival

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kinase cascades [64]. Finally, radiation therapy contributes to a more insidious pattern of endothelial damage, microvascular dysfunction, and progressive fibrosis that may culminate in ischemic or restrictive phenotypes [65]. Together, these therapies highlight the diverse yet intersecting mechanisms that define the CTRCD continuum.

Beyond individual agents, synergistic cardiotoxicity emerges with modern combination regimens. For example, there is an increased incidence of CTRCD in patients treated with both anthracyclines and trastuzumab, with preclinical data suggesting that HER2 blockade increases susceptibility to anthracycline-induced injury [66, 67]. Treatment protocols modeled after KEYNOTE-522, which pair anthracyclines with ICIs in triple-negative breast cancer, demonstrate that cardiac injury may be amplified through oxidative and immune-mediated mechanisms [68]. Regimens combining vascular endothelial growth factor inhibitors with ICIs, commonly used in renal and lung cancers, can compound endothelial dysfunction and promote hypertensive remodeling [69, 70]. Furthermore, combinations of HER2-targeted agents with taxanes or platinum-based cytotoxics, which are now used in esophageal, gastric, and gynecologic malignancies, may exacerbate mitochondrial stress and impair contractile reserve [71]. Although systematic data remain limited, these emerging patterns underscore the importance of monitoring cardiac function and considering potential cardiovascular risks when administering combination regimens.

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Shared Molecular Mechanisms between DbCM and CTRCD

A growing body of evidence suggests that DbCM and CTRCD share several molecular pathways that contribute to cardiac injury [26-34]. At the cellular level, both diabetes and certain cancer therapies promote elevated intracellular ROS levels, leading to oxidative stress secondary to lipid peroxidation, protein damage, and mitochondrial DNA injury [72, 73]. Mitochondrial dysfunction and impaired ATP production are central features of both DbCM and

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CTRCD, contributing to an energy-starved myocardium and contractile dysfunction [74, 75]. Inflammation and cytokine signaling in diabetes mirrors inflammatory cascades activated by certain cancer therapies and lead to endothelial dysfunction and myocardial fibrosis, e.g., ICIs (Table 1) [76, 77]. Both DbCM and CTRCD demonstrate myocardial interstitial fibrosis and extracellular matrix remodeling due to enhanced collagen and fibronectin deposition in the interstitium predisposing to stiffening of the ventricular wall and impaired compliance [78, 79]. Abnormalities in calcium handling are a hallmark of both diabetes and chemotherapy-related cardiac injury. Impaired sarcoplasmic reticulum calcium uptake reduced sarcoplasmic/endoplasmic Reticulum Ca2+-ATPase 2a (SERCA2a) activity, and increased intracellular calcium contribute to impaired relaxation and arrhythmogenesis. Autophagy, apoptosis, and disruption of normal autophagic flux and increased cardiomyocyte apoptosis are seen in both DbCM and CTRCD [80].

Beyond these canonical pathways, adipose-cardiac crosstalk has emerged as a unifying driver of metabolic cardiomyopathy and a plausible amplifier of cancer-therapy susceptibility. In states of visceral adiposity, expansion and dysfunction of adipose tissue remodel the secretome toward a pro-inflammatory, pro-fibrotic adipokine profile (increases in levels of leptin, resistin, RBP4; relative decreases in adiponectin) that promotes endothelial dysfunction, sodium retention, microvascular inflammation, myocardial steatosis, and interstitial fibrosis: features shared by DbCM and prevalent in HF with preserved ejection fraction (HFpEF) phenotypes [81-85]. The “adipokine hypothesis” reframes diabetes-associated cardiomyopathy as a disease of adipose signaling rather than hyperglycemia, with adiposity functioning as a modifiable risk factor that shapes myocardial biology and treatment response [81,82]. These insights intersect with cardio-oncology because adipokine-driven metaflammation and lipotoxic stress can plausibly lower the threshold for CTRCD, particularly in patients with obesity and diabetes receiving cardiotoxic regimens [83-85].

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Closely linked to adipokine signaling is innate immune activation. Pattern-recognition receptors, especially toll-like receptor 2/toll-like receptor 4 (TLR2/TLR4), sense damage- and pathogen-associated molecular patterns arising from hyperglycemia, lipid peroxidation, and mitochondrial injury; they prime NF-κB-dependent transcription of pro-interleukin-1 beta and nucleotide-binding and leucine-rich repeat-containing receptor (NLR) family pyrin domain containing 3 (NLRP3) components in cardiomyocytes and immune cells [86-88]. Anthracyclines and other antineoplastic agents add a second hit, mitochondrial ROS, iron dysregulation, and damage-associated molecular patterns release, that intensifies innate immune signaling and maladaptive remodeling [88]. Thus, DbCM and CTRCD share two-signal logic: metabolic priming (TLR-NF-κB) plus therapy-induced danger cues.

At the convergence of these pathways lies the NLRP3 inflammasome, increasingly recognized in both DbCM and cardio-oncology. NLRP3 activation (via ROS-thioredoxin-interacting protein (TXNIP), ion flux, or mitochondrial DNA) promotes caspase-1 activation, interleukin-1 beta/interleukin-18 maturation, and pyroptosis, driving fibrosis and systolic/diastolic dysfunction [89-91]. Preclinical and translational reports implicate doxorubicin in triggering NLRP3 signaling in the heart, while metabolic disease provides the primed background [88,89]. In this primed milieu, convergent adipose-cardiac and innate immune pathways create a self-reinforcing loop that lowers the threshold for cardiotoxic remodeling. The NLRP3 inflammasome links metabolic stress to caspase-1/gasdermin-D-mediated pyroptosis and Interleukin-1 beta/Interleukin-18 (IL-1β/IL-18) signaling in the heart [92]. At the same time, epicardial adipose tissue (EAT), a depot that shares microcirculation with the myocardium, undergoes expansion and inflammatory remodeling. Activated EAT releases paracrine and vasocrine mediators that drive microvascular inflammation, fibroblast activation, and diastolic stiffening, producing an HFpEF-like phenotype [93]. Finally, EAT-derived extracellular vesicles and microRNAs provide a mechanistic conduit through which adiposity reprograms myocardial redox balance and matrix pathways [94]. In parallel, trained innate immune memory at hematopoietic and tissue-resident

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macrophage levels sustains NF-κB/inflammasome tone, amplifying injury responses once anthracyclines or other cardiotoxic regimens are introduced [95]. Together, these axes (inflammasome-pyroptosis, EAT-EV signaling, and innate immune reprogramming) offer a contemporary framework for why metabolic comorbidity magnifies susceptibility to CTRCD and suggest intermediate phenotypes (EAT burden, IL-1 family cytokines, EV-miRNA signatures) for risk stratification [92-96].

- DbCM is characterized by diastolic dysfunction, fibrosis, lipid accumulation, and mitochondrial dysfunction driven by hyperglycemia, oxidative stress, and profibrotic signaling.
- CTRCD results from various cancer therapies and involves oxidative stress, mitochondrial injury, immune activation, and fibrotic remodeling.
- DbCM and CTRCD share molecular pathways including increased ROS, mitochondrial dysfunction, inflammation, fibrosis, calcium handling abnormalities, and apoptosis, increasing myocardial vulnerability.

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CLINICAL IMPLICATIONS

A patient with diabetes undergoing cancer therapy may have impaired baseline myocardial reserve, increased propensity toward myocardial injury, and a reduced ability to recover from cardiac insult [11]. These patients are characterized by higher rates of comorbidities and report more severe scores for pain and for poorer general well-being [97]. The additive effect of cancer therapy on an already compromised cardiac substrate necessitates a heightened identification of baseline risk and a carefully tailored surveillance and monitoring strategy. A comprehensive baseline assessment of cardiac function and reserve, through assessments such as transthoracic echocardiography with strain and/or cardiopulmonary stress testing, and

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quantification of myocardial fibrosis through cardiovascular magnetic resonance imaging (CMRI) might help identify subclinical DbCM and stratify patients for early intervention. While not well characterized, baseline assessment for the presence of ischemic coronary disease should be considered as part of risk stratification [98, 99]. However, these strategies remain underutilized. The shared pathophysiology (Figure 1) lends itself to targeted interventions that might mitigate risk.

INSULIN AND INSULIN ANALOGUES

Insulin resistance and hyperinsulinemia contribute to maladaptive myocardial changes such as lipid accumulation, oxidative stress, and mitochondrial dysfunction [100, 101]. Chronic hyperinsulinemia activates the renin-angiotensin system, sympathetic nervous system, and profibrotic signaling, and animal studies have shown that insulin can exacerbate myocardial fibrosis and impair mitochondrial bioenergetics [102, 103]. Insulin exerts its effects through phosphatidylinositol 3-kinase/protein kinase B (PI3K/Akt) and rat sarcoma/rapidly accelerated fibrosarcoma/mitogen-activated protein kinase kinase/extracellular signal-regulated kinase (Ras/Raf/MEK/ERK) pathways. While the former promotes metabolic functions, the latter may drive mitogenic and growth-promoting signals, raising concerns about oncogenic potential with high insulin exposure (See Table 1) [104].

Epidemiologic studies have reported increased cancer incidence-particularly breast, colorectal, and pancreatic-in patients using insulin, especially long-acting analogues like insulin glargine. These findings are thought to stem from insulin's ability to stimulate insulin-like growth factor 1 (IGF-1) pathways, which support tumor growth [105-109]. However, many studies are limited by confounding factors such as disease severity, and results remain mixed. Despite the lack of

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definitive evidence, concerns about mitogenic effects in insulin-treated cancer patients have prompted ongoing investigation into the safest insulin regimens for this population [110].

Insulin Interactions with Cancer Therapies

Insulin may interfere with cancer therapy by activating pathways such as PI3K/Akt, potentially reducing the efficacy of targeted treatments like PI3K or mammalian target of rapamycin (mTOR) inhibitors [111,112]. Hyperinsulinemia has been demonstrated to independently enhance tumor growth and significantly increase cancer mortality risk, even in nonobese individuals without diabetes [113]. It may also increase susceptibility to CTRCD through effects like ROS amplification, fibrosis, endothelial dysfunction, mitochondrial dysfunction, metabolic dysfunction, and pro-inflammatory signaling [114-119]. Though insulin-treated patients show higher cardiovascular risk and mortality, this may reflect more severe underlying disease rather than a direct causal effect [7, 120].

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METFORMIN

Metformin is a first-line agent for the treatment of type 2 diabetes [121]. Beyond glucose lowering, metformin has garnered interest for its pleiotropic benefits, including potential to reduce cancer risk, attenuate cardiovascular disease progression, and modulate key molecular pathways involved in DbCM and CTRCD (See Table 1) [122]. It inhibits hepatic gluconeogenesis and improves peripheral insulin sensitivity, lowering plasma glucose without causing hypoglycemia. On a molecular level, metformin activates adenosine monophosphate (AMP)-activated protein kinase (AMPK), which regulates cellular metabolism, mitochondrial biogenesis, autophagy, and oxidative stress. AMPK activation also inhibits the mTOR pathway, potentially limiting tumor progression by reducing proliferation, angiogenesis, and resistance to apoptosis [123-125]. This activation enhances endothelial function, promotes fatty acid

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oxidation, and decreases fibrosis and inflammation-effects that help preserve myocardial energy balance during physiologic stress such as cancer therapy [126-128].

Metformin may function as a cancer therapy by exerting both direct anti-tumor effects on cancer cells and indirect systemic effects through modulation of glucose and insulin metabolism, ultimately reducing cancer risk and mortality [129]. However, results from clinical trials remain mixed, with some early-phase studies suggesting benefit, while larger randomized trials have failed to consistently confirm these findings [130, 131]. In parallel, metformin has demonstrated multiple mechanisms of cardiac protection in both animal and human studies, including reduced myocardial fibrosis, improved diastolic function, enhanced mitochondrial efficiency, and decreased oxidative stress [132, 133]. Its use has been associated with improved ventricular function and reduced HF hospitalizations in patients with diabetes, and it has shown potential to reverse features of metabolic cardiomyopathy in preclinical models [134].

Metformin has emerged as a promising adjuvant to conventional chemotherapeutic regimens, with evidence suggesting it can enhance treatment efficacy, overcome resistance, and reduce required drug doses, making its interaction with cancer therapies an important area of investigation. [135] Metformin may enhance the effectiveness of chemotherapy and radiation by improving pathological response rates, particularly in patients with higher BMI or triple-positive breast cancer histology [136]. Metformin use in patients with diabetes receiving anthracyclines was associated with a significantly reduced risk of CTRCD and lower mortality, suggesting a protective effect against cardiotoxicity [137]. The ability of metformin to reduce myocardial inflammation, prevent mitochondrial damage, and limit fibrotic remodeling may confer added protection in the presence of cancer therapy-related oxidative stress and metabolic strain [122, 123]. It should be noted that metformin is contraindicated in patients with advanced renal impairment due to the risk of lactic acidosis [138, 139].

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GLUCAGON-LIKE PEPTIDE-1 RECEPTOR AGONISTS

GLP-1 RAs stimulate insulin secretion in a glucose-dependent manner, suppress glucagon release, delay gastric emptying, and reduce appetite [140]. The cardioprotective potential of GLP-1 RAs has gained attention due to their benefits in reducing major adverse cardiovascular events, including HF [141]. GLP-1 RAs may help mitigate the cardiovascular side effects of cancer therapies by reducing oxidative stress, inflammation, apoptosis, and endothelial dysfunction, while supporting autophagy and mitochondrial function [142]. Studies suggest that GLP-1 RAs may help attenuate fibrosis, a key pathologic feature in both DbCM and CTRCD. The role for GLP-1 RAs in CTRCD remains largely unknown [143]. Given the ability of the agents to reduce oxidative stress and improve remodeling, GLP-1 RAs are an attractive therapeutic target for cardiac protection during cancer therapy, though oncology-specific trials are limited [142, 143].

Recent randomized trials in HFpEF extend this rationale and clarify the cardiometabolic-inflammatory mechanisms by which GLP-1 RAs may improve patient-centered outcomes. In STEP-HFpEF, semaglutide produced larger improvements in symptoms and physical limitations, greater gains in exercise capacity, and marked weight reduction, alongside substantial anti-inflammatory effects and superiority on a hierarchical composite [144]. Parallel findings were confirmed in STEP-HFpEF DM, where semaglutide again improved symptoms, exercise capacity, weight, and inflammation, with fewer serious adverse events than placebo [145]. Complementing these studies, the SELECT trial showed that semaglutide 2.4 mg reduced major adverse cardiovascular events in adults with obesity and established atherosclerotic cardiovascular disease who did not have diabetes. Follow-up analyses also found fewer heart-failure-related events, including hospitalizations or urgent visits requiring intravenous therapy [146,147].

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Collectively this data indicates that GLP-1 RAs can modify the HFpEF phenotype by reducing adiposity and systemic inflammation and by improving functional capacity and health status, thereby reshaping the cardiometabolic-inflammatory axis central to HFpEF. Although older liraglutide trials in HFrEF (FIGHT, LIVE) were neutral or raised safety signals, underscoring phenotype-specific effects, the STEP-HFpEF program suggests that in patients with obesity (with or without diabetes), semaglutide favorably affects pathways that overlap with those implicated in CTRCD vulnerability (e.g. low-grade inflammation, endothelial dysfunction, and adverse remodeling) [148, 149]. From a therapeutic-implications standpoint, GLP-1 RAs may be reasonable adjuncts for patients with obesity/diabetes who are undergoing potentially cardiotoxic cancer therapies, not as anti-cancer agents themselves but rather as cardiometabolic conditioners that could reduce symptom burden and inflammatory stress. Emerging dual incretin co-agonists (GLP-1/glucose-dependent insulinotropic polypeptide) share many of the cardiometabolic and anti-inflammatory benefits observed with GLP-1 RAs and may further potentiate these effects through combined incretin pathway activation. Early data suggest similar weight-lowering and metabolic benefits in populations with obesity or diabetes, but cardio-oncology evidence remains preliminary [140, 141, 144-147]. Dedicated cardio-oncology trials are now needed to test these translational hypotheses.

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Dipeptidyl Peptidase-4 Inhibitors

Dipeptidyl peptidase-4 (DPP-4) inhibitors act by inhibiting the DPP-4 enzyme responsible for degrading incretin hormones like GLP-1. By inhibiting DPP-4, they prolong the action of GLP-1, leading to improved insulin secretion and reduced glucagon levels [150, 151]. DPP-4 inhibitors may offer benefits in cancer populations by improving glycemic control while also exerting anti-inflammatory and vascular protective effects, which could help reduce cardiovascular complications during cancer therapy [152]. DPP-4 inhibitors suppress inflammatory pathways such as NF-κB, mitogen-activated protein kinase (MAPK), and cytokine signaling, lowering C-

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reactive protein (CRP) and other mediators, thereby targeting mechanisms like oxidative stress, endothelial dysfunction, and fibrosis that contribute to both DbCM and CTRCD [153]. While they may offer benefits in terms of endothelial protection and glycemic control, their direct impact on cardiotoxicity, especially in the context of cancer therapy, has not been sufficiently investigated (See Table 2 for potential interactions and concerns when combining diabetes medications with specific cancer therapies).

Moreover, DPP-4 functions as an adipokine released predominantly from visceral adipocytes, where it promotes insulin resistance and low-grade inflammation: linking adipose dysfunction to cardiometabolic injury [154]. Clinically, sitagliptin has been associated with reduction in EAT thickness over 24 weeks in adults with T2D on metformin, with EAT change correlating with visceral fat rather than hemoglobin A1c (HbA1c) [155]. Complementing this, the randomized ASSET trial found comparable (no significant between-group differences) in pericardial/epicardial fat and myocardial triglyceride content over 12 weeks with sitagliptin versus empagliflozin by MRI, situating DPP-4 inhibition squarely within the adiposity-cardiac fat axis [156]. Together, these data support positioning DPP-4 inhibitors within an adiposity-centered framework (adipokine signaling, EAT phenotypes) relevant to cardiometabolic vulnerability.

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SGLT2 INHIBITORS

SGLT2is (e.g., empagliflozin, dapagliflozin, canagliflozin) reduce renal glucose and sodium reabsorption, producing glycosuria and natriuresis with modest blood pressure and weight reductions. Beyond hemodynamics, SGLT2 inhibition favorably remodels myocardial energetics (increasing ketone utilization and mitochondrial redox capacity), attenuates oxidative stress, and exerts anti-inflammatory and antifibrotic effects that may be relevant to CTRCD [157-161]. Mechanistically, benefits are linked to shifts in substrate utilization toward ketone oxidation, reduced intracellular sodium and calcium (via putative Na^+/H^+ exchanger-1 modulation),

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AMPK/Sirtuin 1 (SIRT1)/Peroxisome proliferator-activated receptor-γ coactivator 1-α (PGC-1α) signaling, and suppression of NLRP3 inflammasome pathways [158-161].

Across HF phenotypes (reduced and preserved ejection fraction) and regardless of diabetes status, SGLT2is reduce HF hospitalization and improve patient-reported quality of life and functional capacity, while also providing kidney protection [162-164]. These class effects and their pleiotropic mechanisms make SGLT2is foundational agents in cardio-oncology when tolerated, particularly in patients at risk for CTRCD or with established HF.

Preclinical and translational studies demonstrate that SGLT2is limit anthracycline cardiotoxicity by preserving mitochondrial function, reducing oxidative stress and apoptosis, and improving myocardial energetics [163]. Large-animal data further show empagliflozin prevents anthracycline-induced LV dysfunction on serial cardiovascular MRI [164]. Clinically, retrospective cohorts in patients with diabetes and cancer report lower risks of incident CTRCD, HF hospitalizations, and all-cause mortality with SGLT2i use compared with matched controls receiving potentially cardiotoxic therapy [165-167]. A prospective case-control pilot, Use of EMPAgliflozin in the prevention of CARDiotoxicity (EMPACARD-PILOT), found empagliflozin, started one week before anthracyclines in high-risk patients, was associated with a reduced incidence of CTRCD and smaller declines in global longitudinal strain/left ventricular ejection fraction without safety signals, supporting evaluation in randomized trials [168,169].

Current meta-analyses of randomized trials do not show an overall increase in incident cancer with SGLT2is; no consistent association with breast or bladder cancer has been observed. A possible signal for renal cancer has appeared in some analyses and pharmacovigilance datasets, warranting vigilance and longer follow-up [170, 171]. Emerging observational work in breast cancer suggests potential mortality benefits of SGLT2is vs DPP-4 inhibitors, but data remain hypothesis-generating [171].

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SGLT2is also map onto the adiposity-cardiac fat framework: a 2024 network meta-analysis found SGLT2is reduced EAT thickness more than placebo and other therapies in patients with dysmetabolism [172]. Human imaging corroborates this, with dapagliflozin lowering EAT thickness and epicardial glucose uptake over follow-up in type 2 diabetes [173]. Mechanistically, SGLT2is attenuate NLRP3 inflammasome activity and downstream IL-1β/IL-18 signaling in cardiometabolic tissues, aligning their effects with adiposity-driven inflammation rather than glucose alone [174]. Together, these data situate SGLT2 inhibition within an adiposity/inflammasome paradigm that is relevant to myocardial vulnerability.

Peri-operative Management of Antidiabetic Therapy in Cancer Patients

As part of oncologic management, patients may undergo surgery, making perioperative management of antidiabetic medications essential to mitigate metabolic and cardiovascular risks. Oral agents such as metformin should be withheld at least 24 hours prior to surgery and for 48 hours postoperatively in patients with renal impairment to reduce the risk of lactic acidosis, while SGLT2is should be discontinued at least 72 hours before surgery due to the risk of euglycemic ketoacidosis. Insulin regimens require careful adjustment with frequent glucose monitoring to maintain stable perioperative glycemia and avoid both hypoglycemic and hyperglycemic complications [175].

Hormonal Therapy-Induced Metabolic Dysfunction and Potential Mitigation Strategies

Androgen-deprivation therapy (ADT) has been shown to disrupt metabolic homeostasis by promoting insulin resistance, dyslipidemia, increased adiposity, and loss of lean mass [176]. These changes overlap with mechanisms driving DbCM, suggesting increased vulnerability to cardiotoxicity, particularly in patients with preexisting diabetes or when combined with other cardiotoxic agents. Strategies to prevent or mitigate these adverse effects are therefore of growing clinical interest.

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Antidiabetic agents, such as metformin, SGLT2is, and GLP-1-RAs, have emerged as potential interventions, though clinical evidence is still limited. Studies investigating metformin in patients receiving ADT show modest improvements in metabolic parameters but no clear impact on metabolic syndrome or survival outcomes [177-179]. Currently, the GAIN PC trial is investigating GLP-1 RAs in this population to evaluate whether metabolic modulation can prevent ADT-induced weight gain, insulin resistance, and dyslipidemia [180]. Meanwhile, SGLT2is have shown early promise, with observational data suggesting reduced cardiovascular events in men with prostate cancer and diabetes on hormone therapy [181].

Although these studies were not designed specifically to evaluate cardiotoxicity, they provide early proof-of-concept that metabolic interventions may reduce hormone therapy-associated cardiometabolic risk. Key gaps remain, including long-term cardiovascular outcomes, optimal timing of interventions relative to hormone therapy initiation, and direct measures of myocardial injury or remodeling. These findings support enhanced metabolic monitoring and targeted mitigation strategies to reduce susceptibility to CTRCD in patients receiving ADT.

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MANAGEMENT OF PATIENTS WITH DIABETES UNDERGOING CANCER THERAPY

Lack of guidelines

Currently, there is limited guidance for managing diabetes in patients undergoing cancer therapy, despite their heightened vulnerability to CTRCD. Standardized protocols for risk assessment, therapy selection, and cardiac monitoring in this population are lacking. Enhanced cardiac monitoring, such as echocardiography and biomarkers like high-sensitivity troponin and N-terminal pro-B-type natriuretic peptide (NT-proBNP), is essential, though optimal timing and frequency remain unclear [182]. Interdisciplinary collaboration can support early detection and personalized initiation of cardioprotective therapies.

CLINICAL IMPLICATIONS AND FUTURE DIRECTIONS

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1. Interdisciplinary Collaboration for Tailored Patient Care

Given the shared pathophysiology between DbCM and CTRCD (Figure 1), it is critical to implement interdisciplinary collaboration between oncology, cardiology, endocrinology, and primary care clinicians with the inclusion of a pharmacist. Clinicians must regularly assess cardiac risk factors in cancer patients with diabetes and incorporate early cardiac screening into the oncology care pathway. Intensive multidisciplinary follow-up during and post-therapy should focus on identification of subclinical cardiac dysfunction and provide opportunities for early intervention. (Figure 2)

2. Optimized Pharmacological Management

The increasing recognition of the interplay between diabetes and oncologic therapeutics underscores the need for personalized pharmacological management. When feasible, potentially beneficial therapies such as SGLT2i and metformin should be prioritized over other diabetes agents due to their potential cardioprotective effects (Table 1). For patients at risk of cardiotoxicity, particularly those undergoing chemotherapy or radiation therapy, these medications may mitigate cardiac risk and improve myocardial function. In high-risk populations when feasible, agents with potential cardiac toxicity should be minimized or avoided. Clinical guidelines should be updated to reflect these insights and provide clear recommendations for diabetes management in patients with cancer, integrating cardiovascular safety into treatment plans. Pharmacy colleagues should be regularly engaged in discussions regarding therapeutic strategies.

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3. Early Risk Stratification and Personalized Care Plans

Early risk stratification is critical for managing patients with both diabetes and cancer. Assessing patient history, cardiovascular risk factors, and cancer treatment type can help identify those at highest risk for CTRCD. Risk assessment tools tailored to this population can guide

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personalized care plans, monitoring schedules, and interventions. Routine echocardiography, cardiac biomarkers, and, when appropriate, CMRI, or functional testing should be used to detect subclinical cardiac dysfunction and optimize care.

4. Refining Clinical Guidelines for Managing Diabetes in Cancer Patients

Current clinical guidelines specifically for the management of diabetes and cancer therapy-related cardiotoxicity are lacking. It is imperative that the development of guidelines for pharmacologic management, cardiac monitoring, and therapy adjustments be developed for the diabetic population of patients with cancer is imperative. Existing guidelines from agencies such as the American Diabetes Association (ADA), American Society of Clinical Oncology (ASCO), and International Cardio-Oncology Society could be leveraged to tailor current guidance to this complex population.

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5. Long-term Survivorship and Prevention

For patients with diabetes and cancer, the longitudinal management of cardiovascular health should be a central focus, particularly as these patients transition into cancer survivorship. While post cancer therapy cardiac surveillance is currently underutilized, long-term monitoring of the diabetic population is of particular importance given their high risk for cardiac toxicity over time. Lifestyle interventions, including dietary counseling, physical activity, and structured exercise programs should be incorporated into survivorship programs to mitigate cardiovascular risk. The implementation of patient education programs focused on healthy lifestyle choices for post-cancer treatment is essential to reducing HF and other cardiovascular complications in this complex and vulnerable population.

- There is a lack of standardized guidelines for managing diabetes in cancer patients, necessitating improved cardiac risk assessment, monitoring, and interdisciplinary care.

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- Early risk stratification and personalized care plans, including prioritization of cardioprotective agents like SGLT2i and metformin, are essential to mitigate CTRCD risk.
- Long-term survivorship care with ongoing cardiac surveillance, lifestyle interventions, and patient education is critical to reduce cardiovascular complications in this high-risk population.

Conclusions — Toward Metabolic Cardioprotection in Cancer Care

Diabetes and cancer converge at the myocardial level through shared pathways of injury, including oxidative stress, myocardial dysfunction, and interstitial fibrosis. The complexity of these interactions is further amplified by adiposity-driven and innate-immune circuits (EAT expansion, inflammasome activation, trained immunity). Within this framework, cardio-oncology care must extend beyond the avoidance of or attention to cardiotoxic cancer therapies with instead a focus on the cardiometabolic milieu. Contemporary glucose-lowering therapies differ meaningfully in this regard. Metformin, SGLT2is, and GLP-1 RAs tend to favor myocardial energetics, vascular health, weight loss, and inflammatory tone, whereas insulin and sulfonylureas may, in specific contexts, augment metabolic stress and require careful titration. Early risk stratification, deliberate medication selection, and coordinated management among oncology, cardiology, endocrinology, pharmacy, and primary care can protect cardiac function without compromising cancer efficacy. The field now needs oncology-specific randomized trials that test cardiometabolic strategies (e.g., GLP-1 RAs/SGLT2is) for prevention and treatment of CTRCD, harmonized cardio-oncology end points, and pragmatic pathways that embed ketone monitoring, volume and renal surveillance, and survivorship follow-up into routine care. Bridging DbCM and CTRCD in this way moves the discipline from reaction to prevention: aligning cancer cure with durable cardiovascular health.

Abbreviations

DbCM: Diabetic Cardiomyopathy

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CTRCD: Cancer Therapy-Related Cardiac Dysfunction

SGLT2i: Sodium-glucose Cotransporter 2 inhibitor

LV: Left Ventricle

SGLT2: Sodium-glucose Cotransporter-2

GLP-1: Glucagon-like Peptide 1

GLP-1 RAs: Glucagon-like Peptide-1 Receptor Agonists

HF: Heart Failure

ROS: Reactive Oxygen Species

TGF-β: Transforming Growth Factor-beta

NF-κB: Nuclear Factor-kappa B

AGEs: Advanced Glycation End Products

HER2: Human Epidermal Growth Factor Receptor 2

ICIs: Immune Checkpoint Inhibitors

PD-L1: Programmed Death-Ligand 1

CTLA-4: Cytotoxic T-Lymphocyte Antigen 4

TKIs: Tyrosine Kinase Inhibitors

SERCA2a: Sarcoplasmic/Endoplasmic Reticulum Ca2+-ATPase 2a

HFpEF: Heart Failure with Preserved Ejection Fraction

TLR2: Toll-like Receptor 2

TLR4: Toll-like Receptor 4

NLR: Nucleotide-binding and Leucine-rich Repeat receptors

NLRP3: NLR family pyrin domain containing 3

TXNIP: Thioredoxin-interacting Protein

IL-1β/IL-18: Interleukin-1 beta/Interleukin-18

EAT: Epicardial Adipose Tissue

CMRI: Cardiovascular Magnetic Resonance Imaging

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PI3K/Akt: Phosphoinositide 3-Kinase/Protein Kinase B

Ras/Raf: Rat Sarcoma/Rapidly Accelerated Fibrosarcoma

Ras/Raf/MEK/ERK: Rat Sarcoma/Rapidly Accelerated Fibrosarcoma/Extracellular Signal-Regulated Kinase

IGF-1: Insulin-like Growth Factor 1

mTOR: Mammalian Target of Rapamycin

AMP: Adenosine Monophosphate

AMPK: AMP-activated protein kinase

DPP-4: Dipeptidyl peptidase-4

MAPK: Mitogen-activated Protein Kinase

CRP: C-reactive protein

HbA1c: Hemoglobin A1c

SIRT1: Sirtuin 1

PGC-1α: Peroxisome proliferator-activated receptor-γ coactivator 1-α

NT-proBNP: N-terminal pro-B-type Natriuretic Peptide

ADA: American Diabetes Association

ASCO: American Society of Clinical Oncology

PD-1: Programmed Cell Death Protein 1

DKA: Diabetic Ketoacidosis

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Declarations

Ethics approval and consent to participate

Not applicable

Consent for publication

Not applicable

Availability of data and materials

Available upon reasonable request.

Clinical trial registration

Not applicable

Competing interests

Javed Butler is a consultant to Abbott, Adaptyx, American Regent, Amgen, Applied Therapeutic, AskBio, Astellas, AstraZeneca, Bayer, Boehringer Ingelheim, Boston Scientific, Bristol Myers

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Squibb, Cardiac Dimension, Cardior, CSL Behring, CVRx, Cytokinetics, Edwards, Element Science, Faraday, Foundry, Innolife, Impulse Dynamics, Imbria, Inventiva, Ionis, Lexicon, Lilly, LivaNova, Janssen, Medtronic, Merck, Occlutech, Owkin, Novartis, Novo Nordisk, Pfizer, Pharmacosmos, Pharmain, Prolaio, Regeneron, Renibus, Roche, Salamandra, Sanofi, SC Pharma, Secretome, Sequana, SQ Innovation, Tenex, Tricog, Ultromics, Vifor, and Zoll.

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Authors' contributions

SF conceptualized the manuscript. SF and GB investigated. SF and GB designed methodology. SF GB and MB managed the project. MB supervised. SF GB CEH and MB aided in visualization. SF and GB wrote the original draft. All reviewed and edited.

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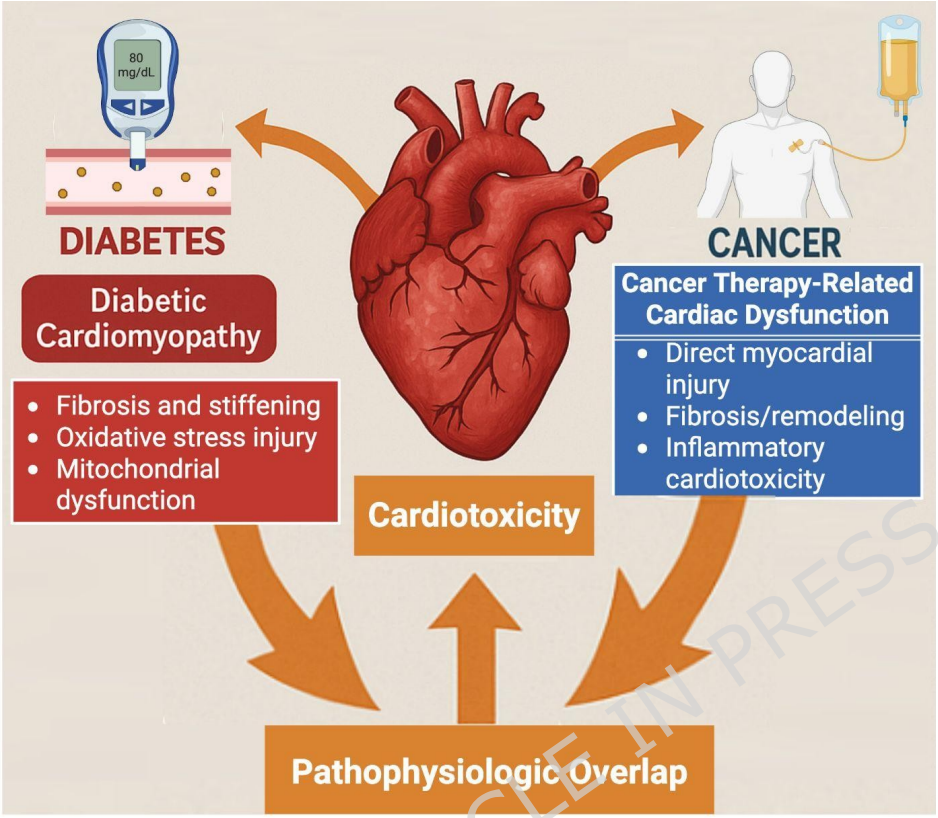
Figure 1. The Complex Interplay Between Diabetes and Cancer



This illustration demonstrates the complex and overlapping pathophysiologic mechanisms linking diabetic cardiomyopathy and cancer therapy-related cardiac dysfunction, additionally highlighting commonly used diabetes therapies and their potential influence of cardiac outcomes in patients also undergoing cancer therapy. The central roles of fibrosis, oxidative stress, and mitochondrial dysfunction are emphasized as shared drivers of myocardial injury. The diagram underscores the importance of these overlapping pathways when managing patients with diabetes undergoing cancer therapy.

Figure 2. Multidisciplinary Strategies for Future Care in Patients with Diabetes and Cancer

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This schematic illustrates a holistic, patient-centered model for managing coexisting diabetes and cancer, emphasizing interdisciplinary collaboration, optimized pharmacologic treatment, early risk stratification, tailored clinical guidelines, and long-term survivorship planning.

Table 1: Mechanisms and Cardiac Implications of Diabetes and Cancer Therapies in the Context of DbCM

Agent	Mechanism of Action	Relevance to DbCM Pathophysiology

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Insulin	Activates PI3K/Akt and MAPK/ERK pathways; promotes glucose uptake and cellular growth [104]	May exacerbate fibrosis, lipid accrual, and oxidative stress; [44,45]
Metformin	Activates AMPK; inhibits hepatic gluconeogenesis and mTOR signaling [123-125]	Reduces myocardial fibrosis, oxidative stress, and improves mitochondrial function; [126-128]
SGLT2 Inhibitors	Inhibits renal glucose reabsorption; reduces preload/afterload; modulates mitochondrial and inflammatory pathways [154-158]	SGLT2 inhibitors reduce heart-failure hospitalization [159-161]
GLP-1 Receptor Agonists	Activates GLP-1 receptors; enhances insulin secretion, delays gastric emptying, reduces appetite [140]	Emerging evidence for anti-inflammatory and cardioprotective effects [146-148]
DPP-4 Inhibitors	Inhibits DPP-4 enzyme; prolongs activity of incretin hormones; increases insulin release, decreases glucagon [150,151]	Limited data may offer mild cardiometabolic benefits.

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Anthracyclines	Intercalates DNA and inhibits topoisomerase II; generates ROS, causing cellular damage [58]	Known to cause irreversible cardiotoxicity through oxidative stress and mitochondrial damage [56,57,59]
Trastuzumab (HER2-targeted therapy)	Inhibits HER2 receptor signaling, disrupting cell proliferation and survival pathways [61]	Reversible cardiotoxicity; interferes with cardiac HER2 survival signaling [60-62]
Radiation Therapy	Damages DNA and promotes fibrosis through ionizing radiation [176]	Leads to fibrosis, vascular damage, and accelerated atherosclerosis [63]
Immune Checkpoint Inhibitors	Blocks PD-1/PD-L1 or CTLA-4 to activate T-cell mediated anti-tumor immunity [177]	Can trigger immune-mediated myocarditis and pericarditis, especially in predisposed individuals [64]
Tyrosine Kinase Inhibitors	Inhibits kinase pathways involved in tumor growth, angiogenesis, and survival [178]	Associated with hypertension, arrhythmia, and ventricular dysfunction [67-71]

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This table summarizes the mechanisms of action of commonly used diabetic therapies and oncologic therapies, highlighting potential implications for DbCM.

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Table 2: Drug–Drug Interaction of Diabetes and Cancer Therapies: Risks and Considerations for Clinical Management

Diabetes Medication	Cancer Therapy / Class	Mechanism and Clinical Implications
Metformin	Tyrosine Kinase Inhibitors (TKIs) (e.g., imatinib, sunitinib)	TKIs may impair renal tubular secretion and reduce metformin clearance, increasing the risk of lactic acidosis if renal function is compromised; renal monitoring is advised [122,123,150].
	Anthracyclines (e.g., doxorubicin)	No direct pharmacokinetic interaction. Metformin may exert cardioprotective and anti-inflammatory effects, potentially mitigating anthracycline-induced cardiotoxicity [116, 117, 121].
Insulin	Immune Checkpoint Inhibitors (ICIs) (e.g., nivolumab, pembrolizumab)	ICIs can trigger immune-mediated β-cell destruction, resulting in new-onset type 1 diabetes or diabetic ketoacidosis (DKA); insulin requirements may fluctuate with systemic inflammation [149].
SGLT2 Inhibitors	Immune Checkpoint Inhibitors (ICIs)	Additive DKA risk due to immune-mediated metabolic stress or reduced oral intake. Monitor for ketosis during acute illness [152].
	Cytotoxic Chemotherapy (e.g., platinum agents, taxanes)	Overlapping risk of volume depletion, renal injury, and euglycemic DKA from chemotherapy-related vomiting or diarrhea. SGLT2 inhibitors

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		should be held during acute illness or dehydration [152].
GLP-1 Receptor Agonists	Tyrosine Kinase Inhibitors (TKIs)	Overlapping gastrointestinal toxicity (nausea, vomiting) can limit tolerance. Monitor hydration and consider dose adjustment if severe symptoms occur [150, 153].
	Anthracyclines / TKIs	Both classes may rarely cause pancreatitis or cardiac dysfunction; concurrent use warrants caution in patients with heart failure or QT interval prolongation [150, 153].

This table outlines potential side effects and/or potential protective effects when using common diabetes medications alongside various cancer therapies. Most diabetes medications have minimal CYP3A4 or P450 metabolism, reducing the likelihood of significant drug-drug interactions with cancer therapies.

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